

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

Percentages

9 questions · ~14 min

03

Q1

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

According to a set of standards, a certain type of substance can contain a maximum of 0.001% phosphorus by mass. If a sample of this substance has a mass of 140 grams, what is the maximum mass, in grams, of phosphorus the sample can contain to meet these standards?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The result of increasing the quantity x by 1,800% is 684. What is the value of x ?

- A. 12,996
- B. 12,312
- C. 38
- D. 36

Q3

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number w is 110% greater than the number z . The number z is 55% less than 50. What is the value of w ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

Jennifer bought a box of Crunchy Grain cereal. The nutrition facts on the box state that a serving size of the cereal is $\frac{3}{4}$ cup and provides 210 calories, 50 of which are calories from fat. In addition, each serving of the cereal provides 180 milligrams of potassium, which is 5% of the daily allowance for adults. If p percent of an adult's daily allowance of potassium is provided by x servings of Crunchy Grain cereal per day, which of the following expresses p in terms of x ?

- A. $p = 0.5x$
- B. $p = 5x$
- C. $p = (0.05)^x$
- D. $p = (1.05)^x$

Q5

PROBLEM-SOLVING AND DATA ANALYSIS

The positive number a is 230% of the number b , and a is 60% of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

- A. 138
- B. 217
- C. 283
- D. 383

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

A scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for 46 days. On February 15, there were 99 dragonflies in the habitat. The percent increase in the number of dragonflies in the habitat from January 1 to February 15 was 12.50%. How many dragonflies were in the habitat on January 1?

- A. 88
- B. 87
- C. 12
- D. 8

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

The value of a collectible comic book increased by 167% from the end of 2011 to the end of 2012 and then decreased by 16% from the end of 2012 to the end of 2013. What was the net percentage increase in the value of the collectible comic book from the end of 2011 to the end of 2013?

- A. 124.28%
- B. 140.28%
- C. 151.00%
- D. 209.72%

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

140 is $p\%$ greater than 10. What is the value of p ?

- A. 1,400
- B. 1,300
- C. 140
- D. 130

Q9

PROBLEM-SOLVING AND DATA ANALYSIS

In 2008, Zinah earned 14% more than in 2007, and in 2009 Zinah earned 4% more than in 2008. If Zinah earned y times as much in 2009 as in 2007, what is the value of y ?

- A. 0.5600
- B. 1.0056
- C. 1.1800
- D. 1.1856